How-To Library

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Configuring the Client for ODBC Connectivity

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Abstract

This article describes how to configure a Windows 7 or Windows Server 2008 client machine for Dynamic Data Masking when the client connects to the database with an ODBC connection.

Supported Versions

- Dynamic Data Masking 9.8.0 or later
- Windows 7
- Windows Server 2008

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Overview

Before you use the Dynamic Data Masking Generic ODBC approach to mask data for a database that uses an ODBC connection, you must configure the client machine.

The Dynamic Data Masking installation includes ODBC DLL files for 64-bit and 32-bit Windows. You must save the files in the Windows system directory and create Dynamic Data Masking host and port environment variables. Optionally, you can create a Data Source in the Windows ODBC Administrator to test the configuration.

1. Verify the requirements for ODBC client configuration.
2. Grant the required permissions to the user that performs the setup for the Windows Driver Manager files.
3. Set up the Dynamic Data Masking Driver Manager proxies.
4. Create the environment variables.
5. Optionally, create the Windows Data Source to test the client configuration.

Step 1. Verify the Requirements

Verify the following requirements before you configure the client:

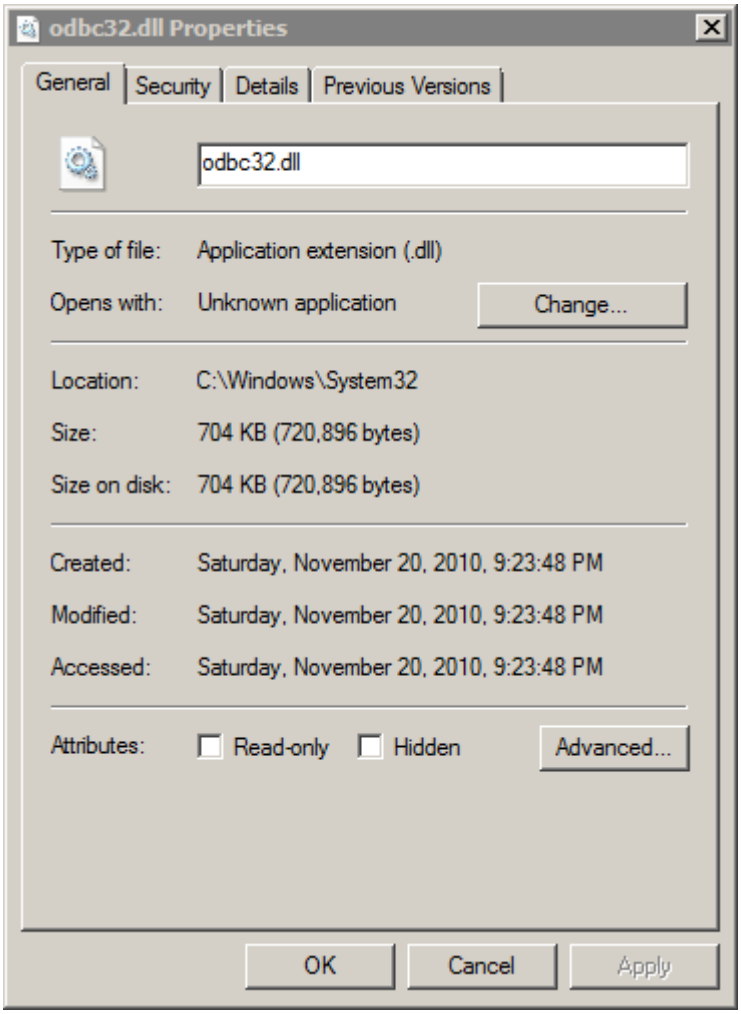
- The client must have a Windows 7 or Windows Server 2008 operating system.
- The Windows user that performs the setup must have administrator privileges to modify the ODBC DLLs or the user must have the rights to grant the required privileges or ownership.
- The Windows user that performs the setup must have the required permissions to rename the Windows ODBC Driver Manager (odbc32.dll).

Step 2. Grant File Permissions

Before you set up the Driver Manager proxies, you must identify the Driver Manager proxy that you want to install based on the target application architecture and based on the ODBC 32-bit or 64-bit usage of the application. Then you must change the owner of the Windows Driver Manager files and grant the required permissions.

1. Find the ODBC Driver Manager DLL file on the Windows machine based on the architecture of the target application. Complete the remaining steps for each of the required Driver Manager files on the machine.
   a. On a 64-bit Windows machine, you can find the following Windows Driver Manager files:
      - 64-bit Driver Manager: `<Windows installation>\System32\odbc32.dll`
      - 32-bit Driver Manager: `<Windows installation>\SysWOW64\odbc32.dll`
   b. On a 32-bit Windows machine, you can find the following Windows Driver Manager file:
      - 32-bit Driver Manager: `<Windows installation>\System32\odbc32.dll`
2. In Windows Explorer, right-click the file and click **Properties**.

   The **odbc32.dll Properties** window opens. The following image shows the window:



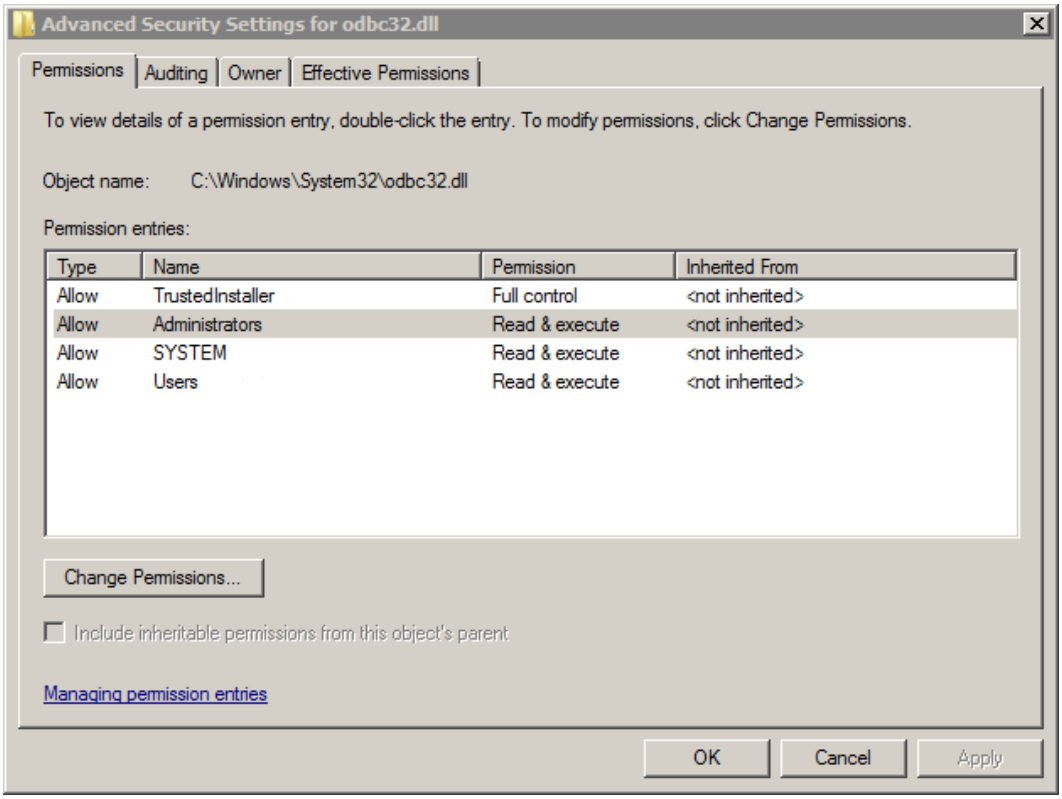
3. Select the **Security** tab and click **Advanced**.

   The **Advanced Security Settings for odbc32.dll** window opens.

4. Select the **Owner** tab and click **Edit**.
5. In the window that opens, select the current user and click **Apply**.
6. Click **OK** to close the window.
7. Click **OK** to close the **Advanced Security Settings for odbc32.dll** window.
8. In the **odbc32.dll Properties** window, click **OK** to close the window. You must close the properties window before you make additional changes to the file.
9. Reopen the **odbc32.dll Properties** window. Right-click the file and click **Properties**.
10. Select the **Security** tab and click **Advanced**.

    The **Advanced Security Settings for odbc32.dll** window opens.
11. In the **Permissions** tab, select the Administrators group and click **Change Permissions**.

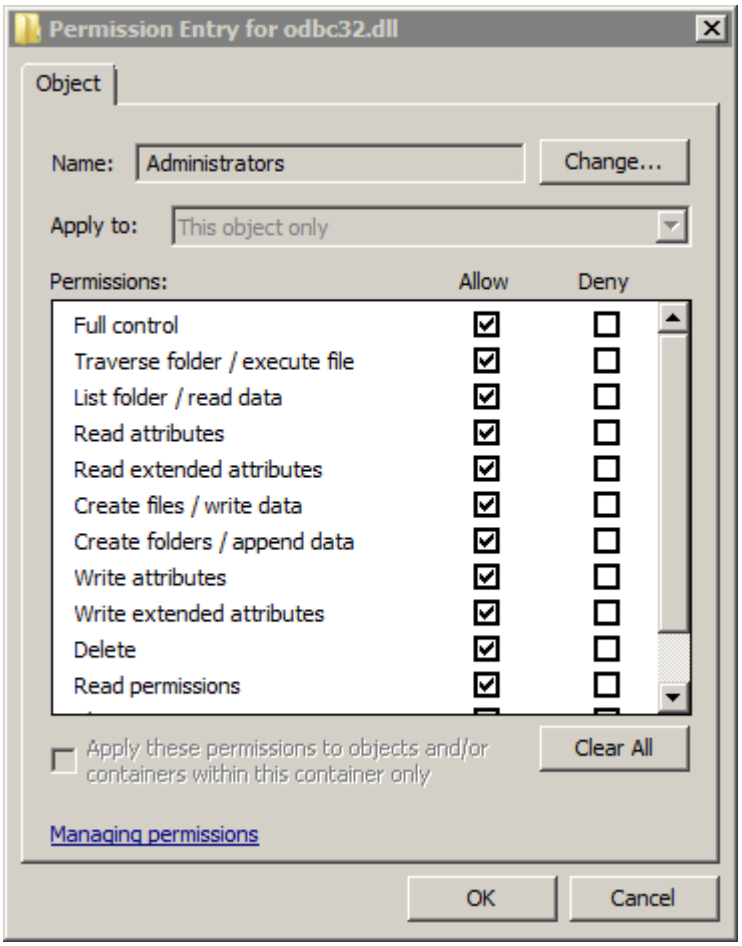
    The following image shows the **Permissions** tab:



12. In the window that opens, select the Administrators group again and click **Edit**.

    The **Permission Entry for odbc32.dll** window opens.
13. Click the **Allow** box in the **Full control** row to grant full permissions to the user.

    The following image shows the window with the **Full control** box selected:



14. Click **OK** to close the window.
15. Click **OK** to close the previous window. A dialog box asks if you want to continue. Click **Yes** to close the window.
16. Click **OK** to close the **Advanced Security Settings for odbc32.dll** window.
17. Click **OK** again to close the **odbc32.dll Properties** window.

    The permissions changes are saved.

Step 3. Set Up the Driver Manager Proxies

Save the Dynamic Data Masking Generic ODBC DLL files in the Windows system directory.

Complete the following steps for each Windows Driver Manager file that you edited in the previous step.

1. Rename the Windows Driver Manager file to odbc32.dll. Rename `odbc32.dll` to `odbc32o.dll`.

2. Find the Dynamic Data Masking Driver Manager proxy based on the architecture of the target client application.
   a. Find the following Dynamic Data Masking Generic ODBC DLL file for a 64-bit application:
      - `<Dynamic Data Masking installation>\Wrappers\odbc\Windows\odbc64\GenericOdbc64.dll`
   b. Find the following Dynamic Data Masking Generic ODBC DLL file for a 32-bit application:
      - `<Dynamic Data Masking installation>\Wrappers\odbc\Windows\odbc32\GenericOdbc32.dll`
3. Copy the Dynamic Data Masking Driver Manger file to the Windows system directory.
   a. Copy `GenericOdbc64.dll` to the following directory:

      `<Windows installation>\System32`
   b. Copy `GenericOdbc32.dll` to the following directory:
      - On a 64-bit machine: `<Windows installation>\SysWOW64`
      - On a 32-bit machine: `<Windows installation>\System32`
4. Rename the Dynamic Data Masking Generic ODBC DLL file to `odbc32.dll` for the 64-bit and 32-bit version.
   a. Rename `GenericOdbc64.dll` to `odbc32.dll`.
   b. Rename `GenericOdbc32.dll` to `odbc32.dll`.

Step 4. Create the Environment Variables

Create the Dynamic Data Masking host environment variables.

1. Open the Windows Start menu, right click **Computer**, and click **Properties**.

   The **Control Panel** opens.
2. On the right side of the Control Panel, click **Advanced system settings**.

   The **System Properties** window opens.
3. In the **System Properties** window, click the **Advanced** tab and click **Environment Variables** at the bottom of the window.

   The **Environment Variables** window opens.
4. Under the **System variables** box, click **New**.

   The **New System Variable** window opens.
5. In the **New System Variable** window, enter the following properties:

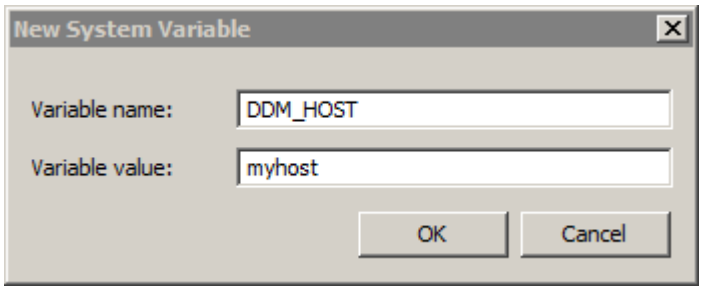
   Variable Name

   DDM_HOST

   Variable Value

   Dynamic Data Masking Server host name.

   The following image shows the **New System Variable** window:



6. Click **OK** to close the window.

   The DDM_HOST environment variable appears in the list of system variables.
7. In the System variables box, click New to add another variable.

   The **New System Variable** window opens.
8. In the **New System Variable** window, enter the following properties:

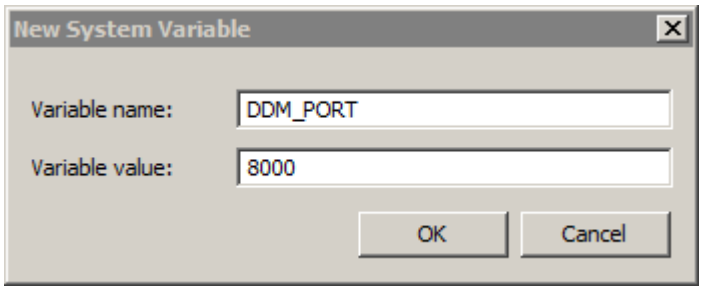
   Variable Name

   DDM_PORT

   Variable Value

   Dynamic Data Masking Generic ODBC service port.

   The following image shows the **New System Variable** window:



9. Click **OK** to close the window.

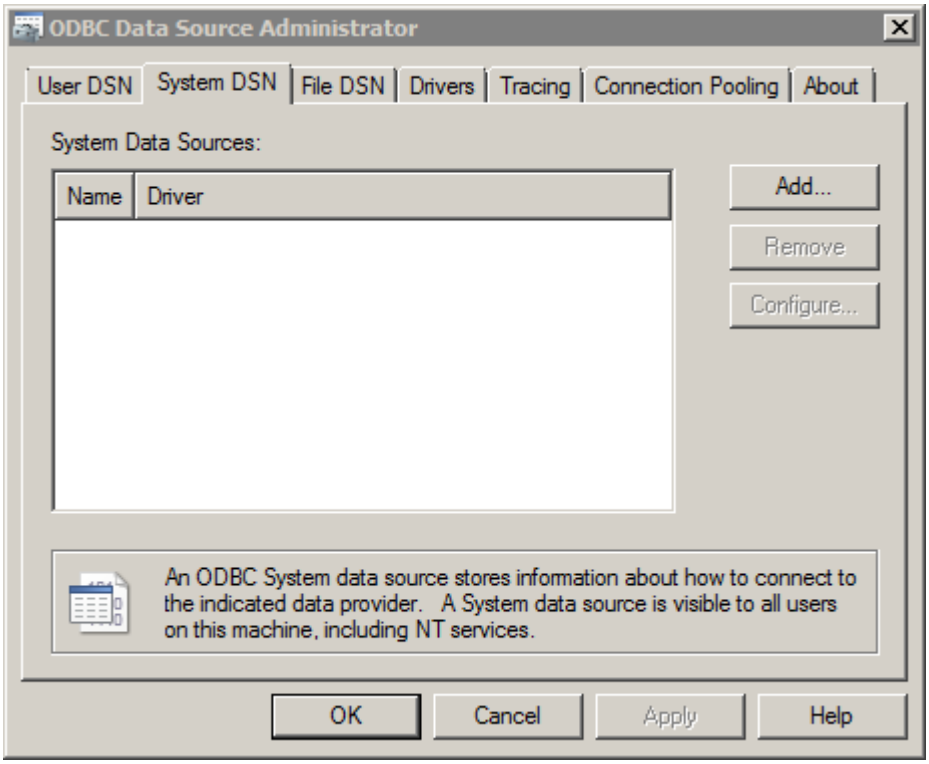
   The DDM_PORT environment variable appears in the list of system variables.
10. Click **OK** to close the **Environment Variables** window.
11. Click **OK** to close the **System Properties** window.

Step 5. Create a Windows Data Source (Test the Setup)

To test the client configuration, or if a System Data Source is not available on the client machine, you can create a System Data Source with the Windows ODBC Data Source Administrator, based on the architecture of the machine.

1. Open the Windows ODBC Data Source Administrator.
   a. On a 64-bit Windows machine, you can find the ODBC Data Source Administrator in the following locations:
      - 64-bit ODBC Administrator: `<Windows installation>\System32\odbcad32.exe`
      - 32-bit ODBC Administrator: `<Windows installation>\SysWOW64\odbcad32.exe`
   b. On a 32-bit Windows machine, you can find the ODBC Data Source Administrator in the following location:
      - 32-bit ODBC Administrator: `<Windows installation>\System32\odbcad32.exe`
2. To view the System Data Sources, click the **System DSN** tab at the top of the window.

   The following image shows the **System DSN** properties:



3. Click **Add**.

   The **Create New Data Source** window opens.
4. In the **Create New Data Source** window, click the name of the ODBC driver that you want to create a DSN for, based on the database that you want to connect to.
5. Click **Finish**.

   The ODBC setup window opens.
6. Provide the DSN details that the ODBC setup window requires and click **Test Connection**. When the successful connection dialog box appears, click **OK** to close the dialog box.

7. Click **OK** to close the ODBC setup window.

   The DSN appears in the **System DSN** tab of the **ODBC Data Source Administrator** window.

8. Note the System Data Source Name property value, which the client application requires to connect to the target database.
9. Click **OK** to close the ODBC Data Source Administrator.

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